

Bernice Animwaa Boateng

berniceboateng775@gmail.com | linkedin.com/in/baboateng | github.com/Berniceboateng775 | [Bernice's portfolio](#)

SUMMARY

Software developer and data analyst with experience building and deploying machine learning applications. I enjoy working on projects where I can combine software development, data and AI to solve real problems. I have worked across the stack, from training models and working with data to building APIs, applications and deploying them. I'm a curious learner who enjoys picking up new technologies and figuring things out along the way.

EXPERIENCE

Software Engineer and Data Analyst

Dec. 2025 – Present

Really Great Tech

Accra, Ghana

SEMLER: Assisted senior data analysts on a project predicting myocardial strain from PPG signals, using XGBoost regression with Leave-One-Clinic-Out validation across clinics

Worked on feature extraction, recursive feature elimination, and loss function tuning, and built pipeline scripts for leakage-safe hyperparameter tuning

AppWork VoiceAI: Assisted on an AI system that detects distress in inbound calls for a property management company by labelling data from 911 calls and building a pipeline that combines audio and text signals

Software Engineer and IT Intern

Apr. 2025 – Aug. 2025

Cocoa Clinic

Accra, Ghana

Assisted with maintaining and updating the hospital's electronic health record (EHR) systems.

Collaborated with the IT team on digital transformation initiatives and internal reporting tools.

Provided technical support for internal software and hardware issues.

PROJECTS

Vellichor AI – Emotion & Theme Analysis | *FastAPI, GPT-2, Hugging Face* [Live App](#) | [Backend](#) | [ML Model](#)

Fine-tuned a custom GPT-2 model end-to-end to classify the emotional vibe and subtext of written dialogue into 7 universal categories

Used context-aware engineering and dropout regularization to outperform standard models on nuanced fiction analysis by reading subtext rather than surface-level keywords

MedMate DDI – Drug Interaction Checker | *Python, React, Django, Machine Learning* [Live App](#) | [ML Model](#)

Built a full-stack healthcare application for detecting drug-drug interactions using custom-trained ML models

Trained transformer-based models on pharmacological datasets to predict interaction severity and generate clinical recommendations

Implemented AI-driven drug substitution to recommend safer alternatives when harmful interactions are detected

Deployed backend on Render (Django REST API) and frontend on Vercel (React) with a PostgreSQL database

CardioGlobal – Heart Disease Data Visualization | *Python, Random Forest, LightGBM* [Live App](#)

Built for healthcare professionals and researchers who need a single tool to understand worldwide cardiovascular disease patterns

Combined WHO/GHO global mortality data with the UCI Cleveland Heart Disease dataset, surfacing three machine learning components in a dedicated ML Insights tab

Book App & Reader Community Platform | *React, Backend APIs, Database Design*

Building a book-centric application that allows users to discover books and connect within a reader-focused social community

Designing features for user interaction, social engagement, and content sharing to foster meaningful connections among readers

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS, SQL

AI/ML: TensorFlow, Hugging Face Transformers, XGBoost, LightGBM, Random Forest, GPT-2 fine-tuning

Frameworks & Databases: React.js, Node.js, Flask, Django, FastAPI, MongoDB, PostgreSQL, Supabase

Tools: Render, Vercel, Git, Docker, Postman, Figma, Microsoft Word, Microsoft Excel

EDUCATION

University of Ghana – B.S. Computer Engineering

Accra, Ghana

Sep. 2021 – Nov. 2025